This report breaks down the performance of five specific trading orders, including our large "parent" orders, the smaller "child" orders they were broken into, the actual trades that happened in the market, and the changing price quotes throughout the day.

Step 1: Gathering and Organizing the Data

First, I gathered all our raw data files. We had four main files:

- **Parent Orders:** The big-picture orders we wanted to execute.
- **Child Orders:** The smaller, bite-sized pieces we actually sent to the market.
- **Trades:** Every single trade that happened in the market that day.
- **Quotes:** The changing "bid" (buy) and "ask" (sell) prices in the market.

Before I could do any math, I had to clean this data up. I combined the separate "date" and "time" columns into a single timestamp for every single record so that I could accurately line up our orders with what was happening in the broader market at that exact split second.

Step 2: Handling the End-of-Day Rush

To make sure our calculations were fair and accurate, I applied a special rule: any market trades recorded after 2:57 PM were grouped together and assigned a time of 3:00 PM. This allowed me to properly measure our "Market on Close" performance.

Step 3: Crunching the Core Numbers

For each of our five parent orders, I looked at the specific time window during which we were trading. I calculated:

- **Notional Value:** The total amount of money involved in the order (in millions of Chinese Yuan).
- **Trading Speed (ADV%):** How much of the total market volume we made up during our trading window. A higher percentage means we were a bigger fish in the pond during that time.
- **Spread:** The average gap between the buying and selling prices in the market while we traded. A wider spread usually means it's harder to get a good price.

Step 4: Comparing to Benchmarks

To know if we did a good job, we need something to compare our prices against. I calculated our cost (in "basis points" or bps, which are just tiny percentages) compared to several benchmarks:

- **Open Price:** The very first price of the day.
- **Arrival Price:** The market price right before we started trading.
- **IVWAP (Interval Volume-Weighted Average Price):** The average price of all trades in the market during our specific time window.
- **Close Price:** The final price of the day.
- **PWPS (Participation Weighted Price):** A simulated benchmark where I calculated what the price would have been if we had smoothly traded exactly 5% of the market volume throughout our window.

Step 5: Understanding Our Fill Tactics

Finally, I looked at *how* we traded:

- **MOO% (Market on Open):** How much of our order was executed right when the market opened.
- **MOC% (Market on Close):** How much was executed right at the end of the day.
- **Passive vs. Aggressive:** I checked our prices against the market quotes at the exact time of our trades. If we paid the "ask" price to buy immediately, we were **Aggressive**. If we waited at the "bid" price for someone to sell to us, we were **Passive**.

After running all the calculations, I exported the results into a final spreadsheet. We looked at five specific parent orders (V001 through V005). Here is the story for each one:

Order V001

- **The Basics:** This order was worth **15.89 million CNY** and accounted for **0.90%** of the market volume while active. The market spread was relatively wide at 8.60 bps.
- **The Performance:** Our average price wasn't great compared to the market open (costing us 30.88 bps) or arrival price. However, we vastly outperformed the closing price by 121.47 bps, meaning the market likely moved against us later in the day, so buying earlier was a good thing!
- **The Tactic:** We were quite aggressive here. **64.53%** of our trades were aggressive, meaning we crossed the spread to get things done quickly, while only 32.29% were passive. We also did about 3.19% of our trading right at the market open.

Order V002

- **The Basics:** A larger order of **23.39 million CNY**, making up **0.79%** of the market. The spread was much tighter here at only 3.98 bps.
- **The Performance:** This order was a big win compared to the arrival price. We beat the arrival price by **25.83 bps**, meaning we got a much better price than what the market was showing right before we started. We slightly underperformed our IVWAP benchmark (by 4.19 bps).
- **The Tactic:** Similar to V001, we were very aggressive (**66.69%**) and didn't use any special open or close auctions.

Order V003

- **The Basics:** This was our smallest order, worth only **4.47 million CNY**. Because it was small, our trading speed was very low (0.35% of the market).
- **The Performance:** The costs here were quite high across the board compared to the Open (69.38 bps) and Arrival (69.38 bps). It seems the market price was running away from us.
- **The Tactic:** We traded almost 10% of this order right at the market open. Our strategy was much more balanced here: **48.27% passive** and **41.79% aggressive**.

Order V004

- **The Basics:** A heavy order worth **28.49 million CNY**. We traded quite fast here, making up **1.82%** of the market volume. The spread was tight (4.06 bps).
- **The Performance:** Excellent execution. We beat both the Open and Arrival prices by **3.77 bps**, and we beat our simulated 5% participation benchmark (PWPS) by nearly 5 bps.
- **The Tactic:** We executed about 3.27% at the market close. Our trading was fairly balanced, with **46.40% passive** and **52.20% aggressive**.

Order V005

- **The Basics:** Our biggest order by far, coming in at **33.14 million CNY**. We were a major player during this window, accounting for **2.27%** of all market volume.
- **The Performance:** The numbers here are wild! We paid a massive 149.28 bps more than the open/arrival prices. However, we beat the IVWAP benchmark by an incredible **17.86 bps** and beat the PWPS benchmark by **93.13 bps**. This tells me the market was likely highly volatile or trending aggressively during this period.
- **The Tactic:** To handle this massive order, we switched our strategy. Instead of being aggressive, we were highly patient. **61.83%** of our trades were **passive**, meaning we sat back and let the market come to us. Only 36.87% was aggressive.

Looking at the data as a whole, a few major themes stand out that can help us trade better in the future.

Size Changes Strategy

When we had smaller to medium orders (like V001 and V002), our trading algorithms favored an aggressive approach—we went out and grabbed the shares we needed, crossing the spread roughly 65% of the time. However, when we had our absolute largest order (V005), the algorithm smartly flipped the script. It became highly passive (over 61%) to avoid scaring the market and pushing prices away from us.

The IVWAP Benchmark is Our Truest Test

Comparing our trades to the Open or Close prices can be misleading because the market can move a lot over a whole day. Our performance against the IVWAP (the average market price during our specific trading window) was much more stable. Across all five orders, our IVWAP costs ranged from roughly -17 bps to +6 bps. This shows that, on average, our algorithms are doing a good job of tracking the actual market conditions while we are actively trading.

Market Impact is Real

For our fastest-trading orders (V004 and V005, where we were 1.8% to 2.2% of the market), we saw vastly different results against the arrival price. This highlights how tricky it is to trade large amounts quickly. Even with a highly passive strategy on V005, the sheer size of the order likely made it difficult to secure prices near where the market started.

Conclusion

To wrap things up, analyzing these five orders has given us a wonderful window into how our automated trading strategies perform in the real world.

By taking millions of rows of raw data and carefully matching our child orders to split-second market quotes, we were able to see exactly when we were being aggressive and when we were being patient.

Overall, the algorithms are behaving logically. They are aggressive when the orders are small and the spread is tight, and they become patient and passive when the orders are massive. While our performance against start-of-day benchmarks varies wildly due to natural market movements, our performance against the trades happening *at the same time* as ours (IVWAP) proves that we are executing efficiently and securing fair prices for our size.

Gemini Pro 3 is used to formalize the reports and refine grammars.